

ALEXANDER TONG

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EDUCATION

Boston University

B.S. Mechanical Engineering, Aerospace Concentration

Expected May 2028

GPA: 3.71/4.00

EXPERIENCE

Boston University Plummer Laboratory

June 2026 - Present

Mechanical Engineering Intern

Boston, MA

- CNC-machined a 6061-T6 shaker adapter on a Haas VF-2; cut stock mass 40.4% from 3.19 to 1.90 kg with a conical underside chamfer, corner fillets, and 12 drilled and tapped holes
- Generated G-code and tolerance stack-ups for ± 0.005 in holes; cut plate-and-fastener cost from a \$1,170 outsourced estimate to \$71.11
- Predicted 0.44 mm fixture deflection at 3 kg and 20.1 Hz using SolidWorks dynamic FEA; analyzed fastener preload and joint failure modes
- Measured 50–120 mN forces independently with an Instron 68SC-5 to investigate physical Ising models and compare with mathematical predictions

Culturon

February 2026 - June 2026

Mechanical Engineering Intern

Sydney, Australia

- Executed the company's first full-scale nanoparticle deposition mapping using SEM, optical microscopy, and ImageJ
- Proposed turbomolecular- and roughing-pump port and chamber-bottom geometry changes using fluid dynamics and deposition evidence to address particle contamination
- Designed Onshape sheet-metal sample racks for reactor-port insertion and adhesive-free mounting across chamber locations

ENGINEERING PROJECTS

Argo Separation and Recovery Subsystem

June 2025 - Present

Co-Lead Engineer, Boston University Rocket Propulsion Group

- Designed redundant recovery for a 135 lb rocket targeting 100,000 ft; achieved 18.7 lbm against a 20 lbm budget and presented four design reviews
- Introduced and dynamically tested Brummel eye splices to replace knots and reduce quick-links; confirmed minimum 1.5 FOS
- Calculated 823 lbf parachute shock and 1,230 lbf snatch loads by synthesizing five research-paper methods
- Developed CO2 hardware after recurring integrated-test failures and parachute damage; completed stress analysis, CAD, drawings, machining, and vacuum tests in a three-person team
- Led recovery test operations: plans, safety protocols, crew assignments, briefings, firing/abort decisions, and post-test analysis
- Built and operated a 352 L, 0.011 atm vacuum chamber using salvaged pumps and stock for deployment-hardware and avionics tests
- Redesigned and machined cable cutters, reducing mass 49.5% with a 2.01 FOS

AIAA Design-Build-Fly Aircraft

June 2025 - December 2025

Wing Structural Lead, Boston University

- Led two wing designs in Fusion 360; produced a 50% lighter XPS prototype and cut build time from three months to one
- Designed a split NACA 2412 wing with 0.52 taper ratio and 5-degree dihedral; led structural and manufacturing reviews and mentored the succeeding lead

Hybrid Rocket Nozzle

September 2024 - November 2024

Lead Engineer, Boston University Rocket Propulsion Group

- Led five members through three design reviews and a two-second hotfire; cut nozzle mass 73% from the initial design with SolidWorks CAD/FEA, calculating a 7.8 FOS under mechanical loads alone

TECHNICAL SKILLS

SolidWorks, Fusion 360, Onshape, FEA, GD&T, Engineering Drawings, CNC & Manual Machining, Composite Layup, Vacuum Systems, Test Engineering, Python, MATLAB